

Akshay Bhat

Bachelor of Engineering
Artificial Intelligence and Machine Learning
B.M.S College Of Engineering, Bengaluru
CGPA: 8.98

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GitHub Profile
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Portfolio Website

TECHNICAL SKILLS

-Languages: C++ | Python | C | JavaScript | SQL | HTML | CSS
-Frameworks: React.js | Next.js | Tailwind | Flask | TensorFlow | Scikit-learn
-Tools: Git | Google Colab | GCP | Jupyter Notebook | MongoDB | MySQL | SQLite | Vercel | Render
-Course Work: Data Structures | Databases | OOP | Computer Architecture | Machine Learning

PERSONAL PROJECTS

- **ContractPulse** | Python, RoBERTa, DistilBERT, Prophet, Flask, Next.js, MongoDB Apr 2026
Fintech SaaS platform for AI-powered legal contract analysis and breach risk forecasting.
 - Built and fine-tuned a **RoBERTa QA model** on **CUAD** to extract financial obligations via a 7-stage pipeline (segmentation → QA inference → NER → normalization), outputting structured JSON with confidence scores.
 - Integrated **DistilBERT NLI** for cross-contract conflict detection and handled full backend API integration for **Groq** LLM-powered clause extraction; product also features **Facebook Prophet** time-to-breach forecasting.
 - Deployed frontend on **Vercel** and backend on **Hugging Face Spaces**, supporting live PDF uploads with **unbounded clause extraction**, covenant monitoring, and automated breach alert workflows.
- **NeoScore** | Python, XGBoost, SHAP, Flask, Next.js, MongoDB Mar 2026
Explainable AI credit scoring platform for thin-file users in the Indian lending ecosystem.
 - Trained an **XGBoost** model on **307K+ records** with **isotonic calibration** to generate probability scores, then distilled its knowledge into a lightweight student model targeting users with **no prior credit history**.
 - Engineered **30+ features** (debt-to-income, EMI burden, income stability) with **monotonic constraints**; achieved **ROC-AUC 0.79** — strong performance on a historically underserved user segment.
 - Integrated **SHAP** explainability and a **Groq LLM** layer for counterfactual recommendations; deployed via **Flask** with **Google OAuth** and session history on **MongoDB**.
- **MoodQuest** | PyTorch, OpenCV, Flask, Next.js, MongoDB, Docker Nov 2025
Real-time facial expression recognition and mental health monitoring platform.
 - Integrated a fine-tuned **EfficientNet-B0** model into a **Flask** backend, building the full inference pipeline with **OpenCV** Haar Cascades for face detection and frame-level emotion classification.
 - Built and deployed the full-stack wellness platform — clinical assessments (**PHQ-9**, **GAD-7**, **PSS**), longitudinal mood dashboards in **Next.js**, **MongoDB** session persistence, with the frontend on **Vercel** and backend on **Hugging Face Spaces**.
- **CareBridgeAI** | Python, RAG, Pinecone, SentenceTransformers, SciSpaCy Sept 2025
AI-powered humanitarian platform for medical consultation and healthcare resource discovery.
 - Engineered a **RAG pipeline** indexing **250,000+ medical dialogues**, using **SentenceTransformers** for dense retrieval and **Pinecone** as the vector store for high-precision semantic search.
 - Integrated **SciSpaCy** for medical entity extraction, enabling symptom-driven semantic queries across a large-scale healthcare dialogue corpus.
- **FabrAIc** | Python, PyTorch, ResNet50, Flask, Pandas Aug 2025
Intelligent visual search engine for wardrobe management and outfit discovery.
 - Built a **Multi-task ResNet50** model on the **DeepFashion dataset** with auxiliary classification heads for categories and attributes, generating robust **2,048-dimensional visual embeddings**.
 - Engineered a **vector similarity search pipeline** indexing **289,000+ images** using cosine similarity for real-time content-based apparel retrieval, integrated with a Flask backend service.

POSITIONS OF RESPONSIBILITY

• Web Developer & UI Designer

AeroBMSCE Club

Sept 2025 – Feb 2026

ACHIEVEMENTS

- Top 6 of 100 teams — AiVerse 2.0 Hackathon
- Top 15 of 125 teams — Phaseshift Tech Fest '25

Oct 2025 — **MoodQuest**
Sept 2025 — **CareBridgeAI**